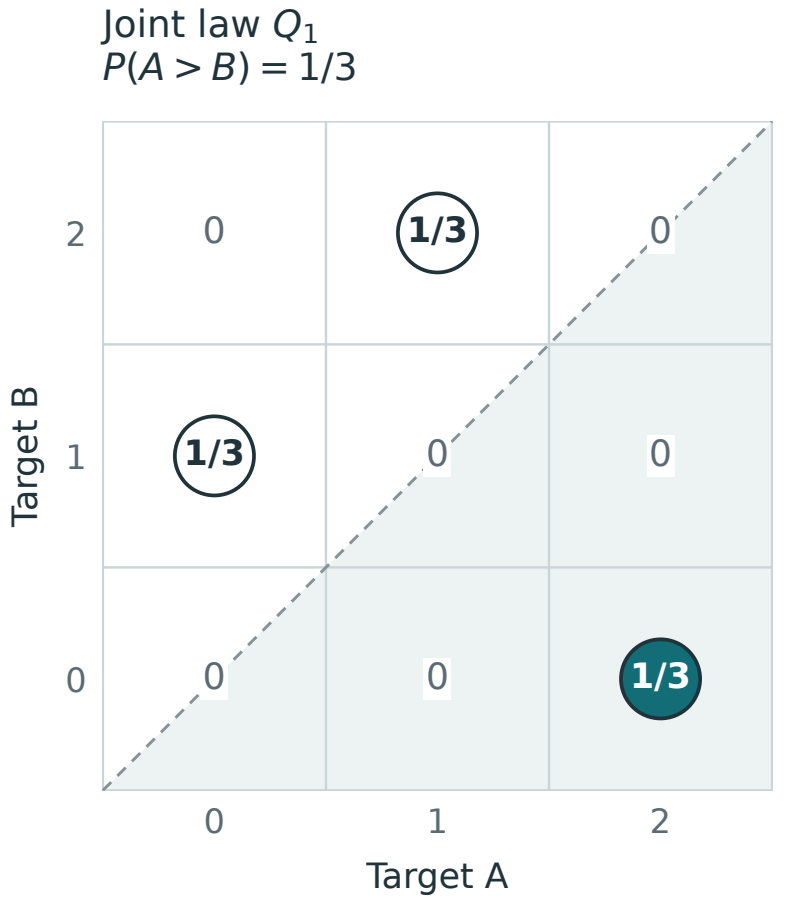
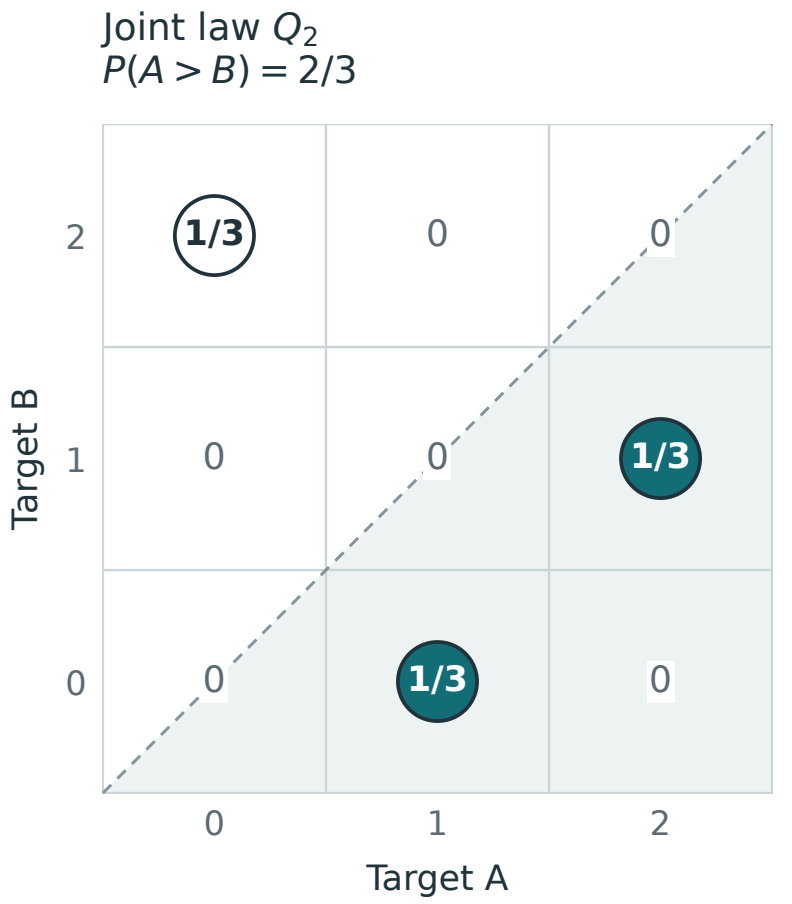
Equal marginals, different comparisons

A and B are each uniform on {0, 1, 2}.
Each occupied cell has mass 1/3.



Every row and column totals 1/3
in both laws.



Shaded half-plane and filled supports: A > B.
Dashed diagonal: A = B. Other cells have mass 0.

Stipulated laws; independence is not assumed.